

ACTIVITY 8: Testing Account Subclasses

Objective

Write a test class that verifies the specialized behavior of SavingsAccount and CurrentAccount.

Requirements

Create a test class `TestAccountSubclasses.java` that:

1. Creates SavingsAccount and CurrentAccount instances
2. Tests savings-specific features (interest calculation)
3. Tests current-specific features (overdraft)
4. Demonstrates polymorphism (treating both as Account)
5. Shows inheritance working properly

Expected Output

When you run your test class, the output should look like this:

text

```
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ACCOUNT SUBCLASSES TEST (SAVINGS & CURRENT)
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>>> Test 1: Creating Accounts

Savings Account: Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No

Current Account: Account #1002 | Jane Smith (30 yrs) | Current | ₹2000.0 | Active | PIN: No

>>> Test 2: Account Type and Minimum Balance

Savings Account - Type: Savings, Minimum Balance: ₹500.0

Current Account - Type: Current, Minimum Balance: ₹1000.0

>>> Test 3: Savings Account - Interest Calculation

Savings Account: Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active

Interest Rate: 4.0% per annum

Interest for 1 year: ₹40.0

Interest for 2 years: ₹80.0

Interest for 5 years: ₹200.0

After 2 years with interest: Balance would be ₹1080.0

>>> Test 4: Current Account - Overdraft Feature

Current Account: Account #1002 | Jane Smith (30 yrs) | Current | ₹2000.0 | Active

Overdraft Limit: ₹5000.0

Available Overdraft: ₹5000.0

Overdraft Used: ₹0.0

Is Using Overdraft: false

Withdrawing ₹1500.0 (goes below minimum balance of ₹1000)

Balance before: ₹2000.0

Withdrawing: ₹1500.0 - SUCCESS

Balance after: ₹500.0

Overdraft Used: ₹500.0

Available Overdraft: ₹4500.0

Is Using Overdraft: true

Attempting to withdraw ₹4000.0 (would exceed overdraft)

Available funds: ₹500.0 (balance) + ₹4500.0 (overdraft) = ₹5000.0

EXCEPTION: Insufficient funds. Available: ₹5000.0 (including ₹5000.0 overdraft), Requested: ₹4000.0

Repaying overdraft of ₹500.0

Balance before repayment: ₹500.0

Overdraft Used before: ₹500.0

Repaying ₹500.0 - SUCCESS

Balance after repayment: ₹1000.0

Overdraft Used after: ₹0.0

Is Using Overdraft: false

>>> Test 5: Polymorphism - Treating Accounts Uniformly

Processing accounts polymorphically:

Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | Type: Savings, Min Balance: ₹500.0

Account #1002 | Jane Smith (30 yrs) | Current | ₹1000.0 | Active | Type: Current, Min Balance: ₹1000.0

Account #1003 | Bob Wilson (35 yrs) | Savings | ₹500.0 | Active | Type: Savings, Min Balance: ₹500.0

Account #1004 | Alice Brown (28 yrs) | Current | ₹1500.0 | Active | Type: Current, Min Balance: ₹1000.0

Total accounts: 4

Total balance across all accounts: ₹4000.0

>>> Test 6: Validation - Invalid Creation Attempts

Attempting to create SavingsAccount with ₹300 (below minimum)

EXCEPTION: Savings account requires minimum balance of ₹500.0. Provided: ₹300.0

Attempting to create CurrentAccount with ₹500 (below minimum)

EXCEPTION: Current account requires minimum balance of ₹1000.0. Provided: ₹500.0

Attempting to create SavingsAccount with age 16

EXCEPTION: Customer must be at least 18 years old. Provided: 16

>>> Test 7: Savings Account - PIN and Operations

Savings Account: Account #1005 | Charlie Green (40 yrs) | Savings | ₹2000.0 | Active

Setting PIN 1234: SUCCESS

Depositing ₹500.0: SUCCESS

Balance after deposit: ₹2500.0

Withdrawing ₹300.0 with correct PIN: SUCCESS

Balance after withdrawal: ₹2200.0

Attempting to withdraw ₹2000.0 (would violate minimum balance)

EXCEPTION: Cannot withdraw. Minimum balance of ₹500.0 required. Available after withdrawal: ₹200.0

>>> Test 8: Current Account - Active Status Operations

Current Account: Account #1006 | Diana Prince (35 yrs) | Current | ₹3000.0 | Active

Closing account: SUCCESS

Attempting to deposit ₹100.0 on closed account

EXCEPTION: Account is inactive. Please reopen the account or contact support.

Reopening account: SUCCESS

Depositing ₹100.0 after reopen: SUCCESS

Balance after deposit: ₹3100.0

>>> Test 9: All Accounts Summary

Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No

Account #1002 | Jane Smith (30 yrs) | Current | ₹1000.0 | Active | PIN: No

Account #1003 | Bob Wilson (35 yrs) | Savings | ₹500.0 | Active | PIN: No

Account #1004 | Alice Brown (28 yrs) | Current | ₹1500.0 | Active | PIN: No

Account #1005 | Charlie Green (40 yrs) | Savings | ₹2200.0 | Active | PIN: Yes

Account #1006 | Diana Prince (35 yrs) | Current | ₹3100.0 | Active | PIN: No

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TEST COMPLETED!

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Task

Write the complete `TestAccountSubclasses` class in a file called `TestAccountSubclasses.java`.
Handle all exceptions and display appropriate messages.
